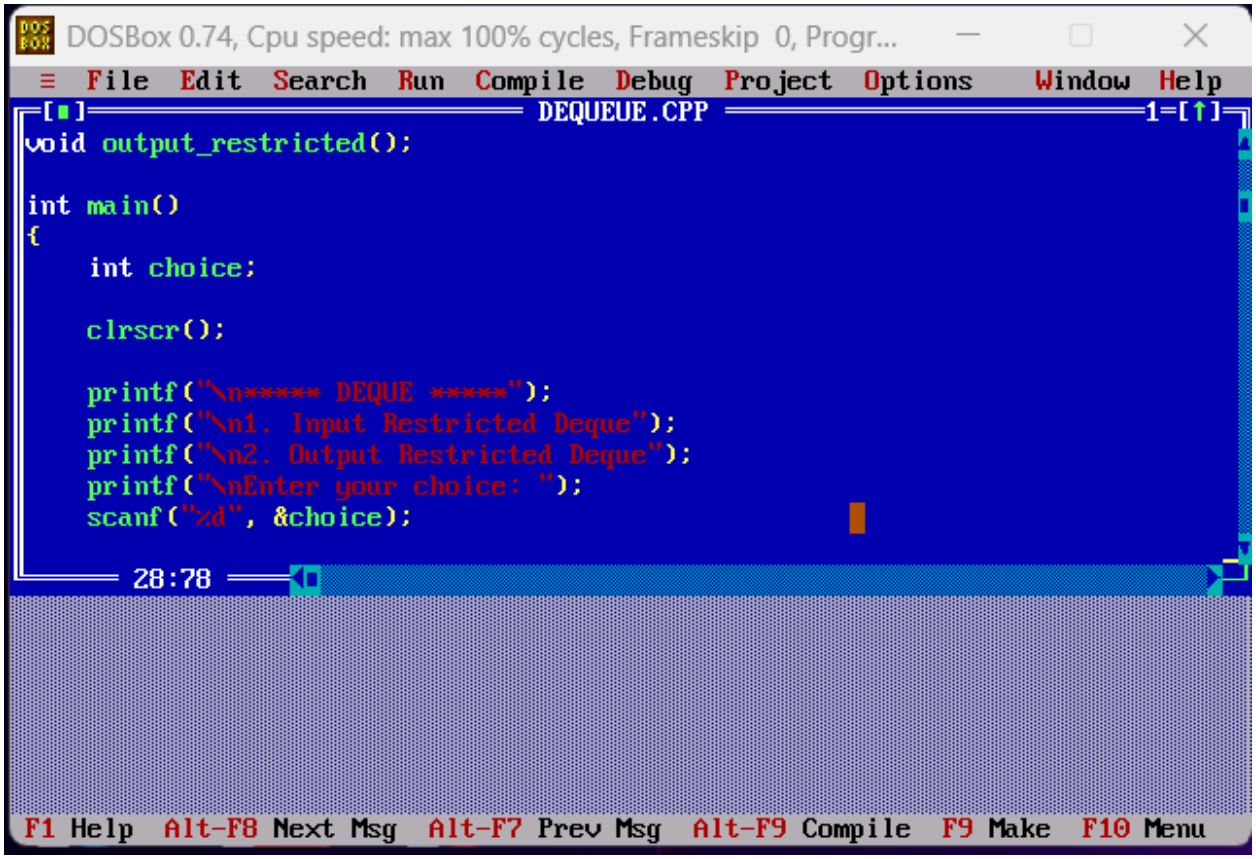
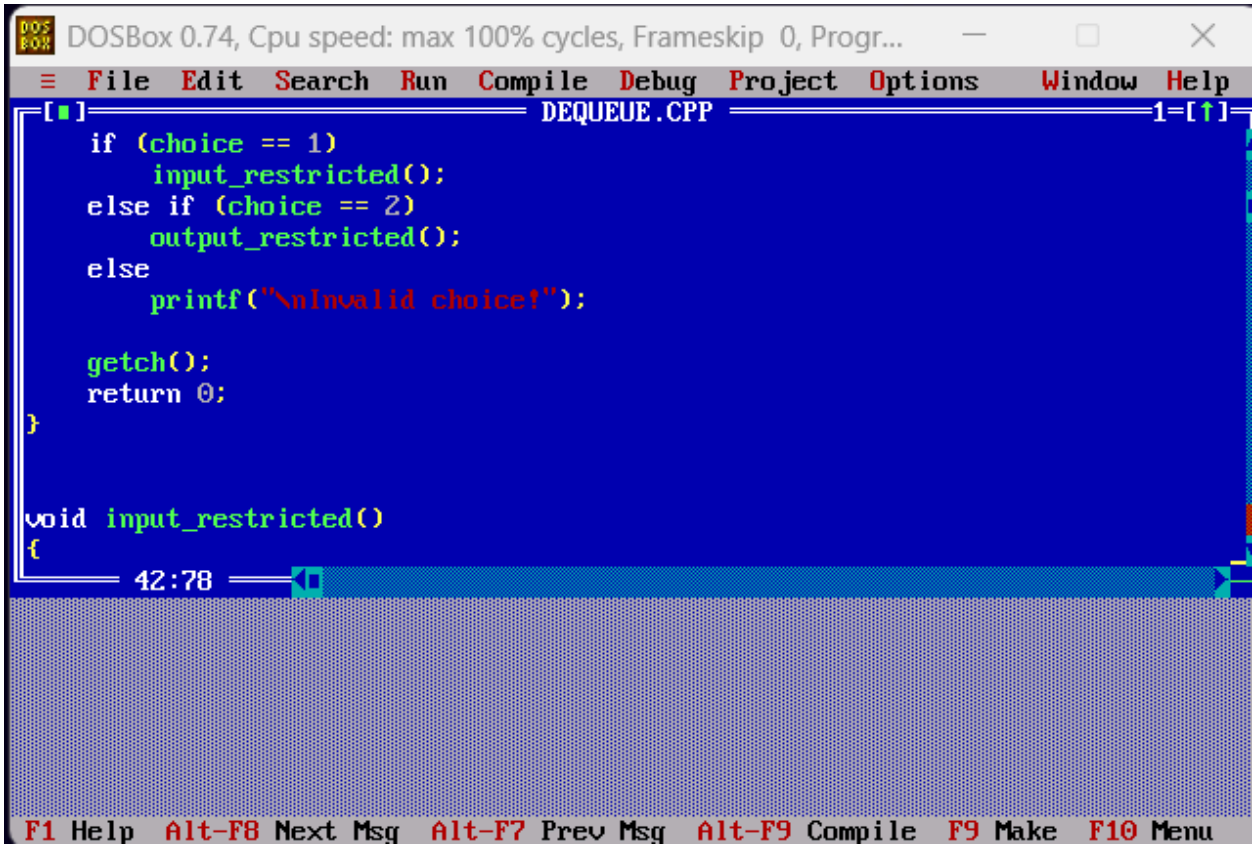
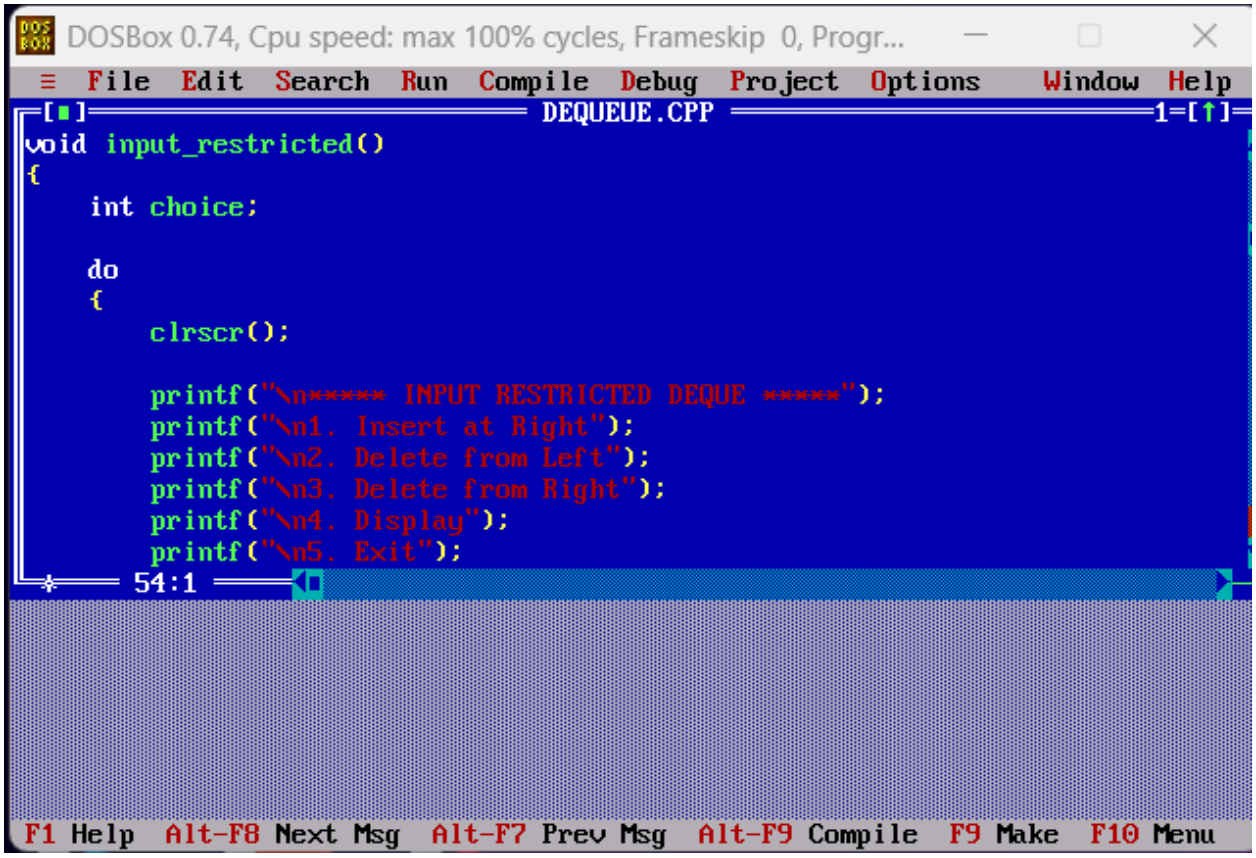
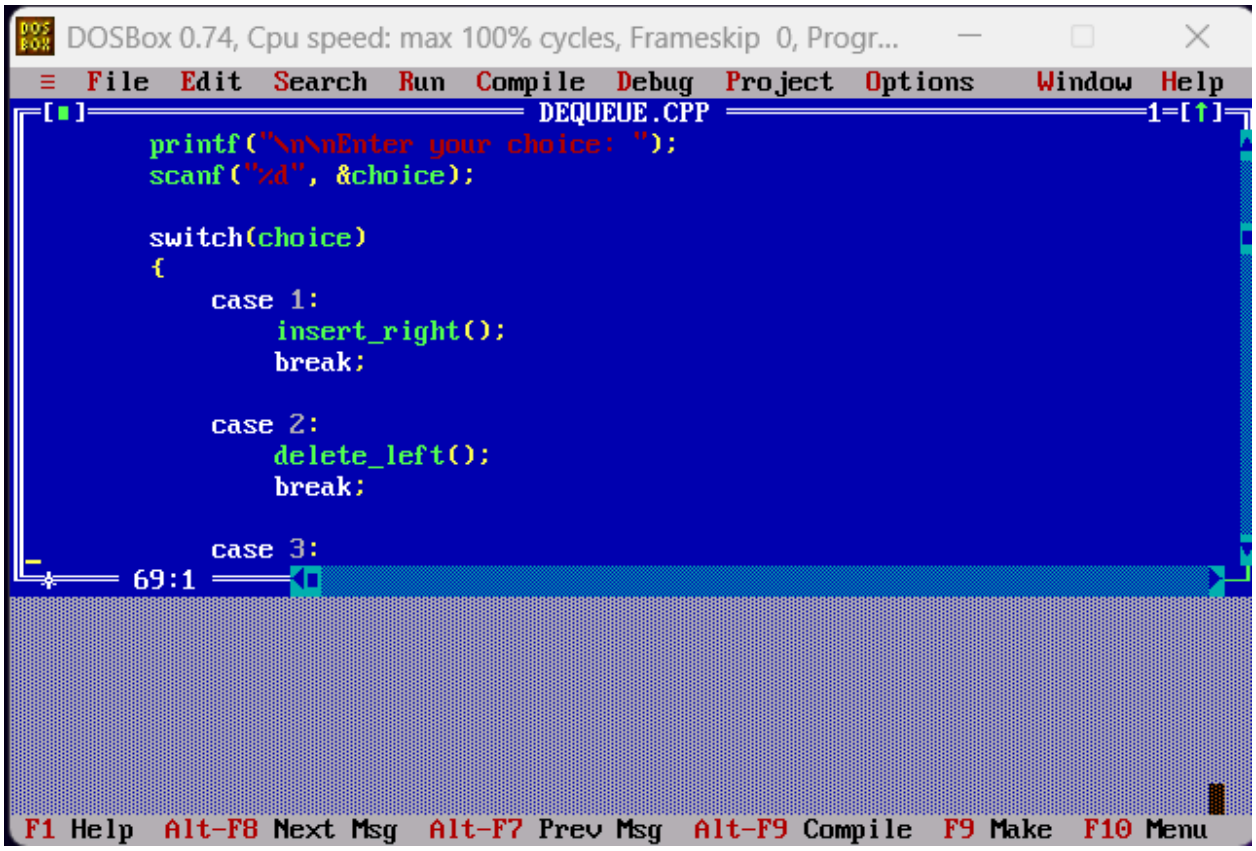
DEQUEUE

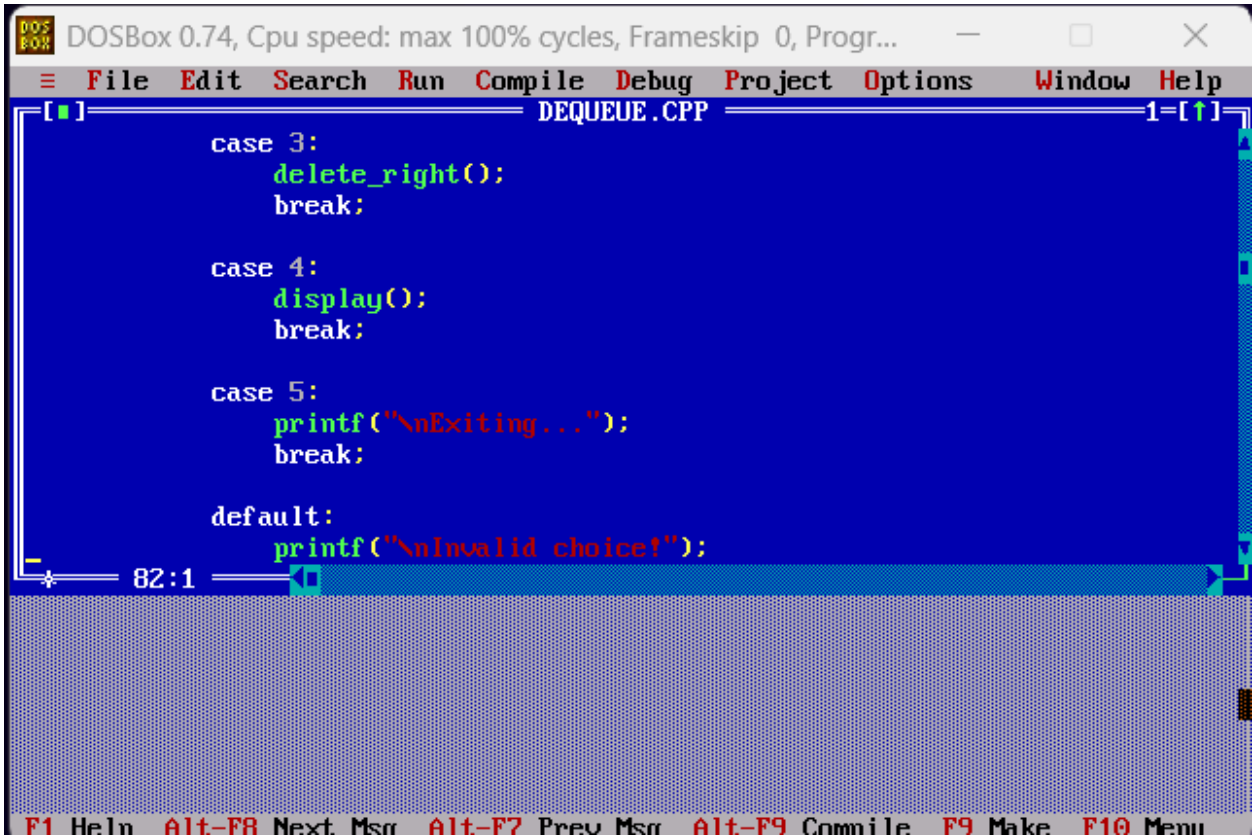
Program:

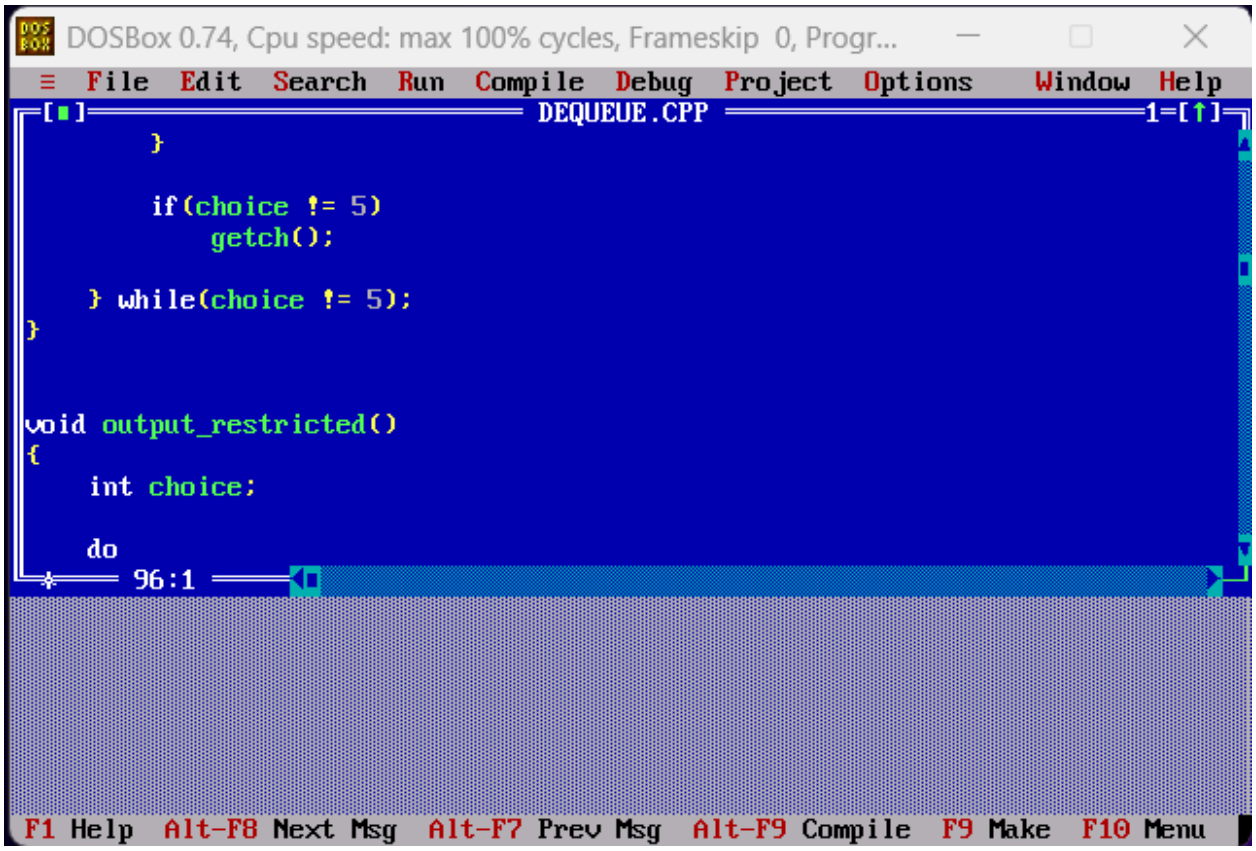

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
File Edit Search Run Compile Debug Project Options Window Help
DEQUEUE.CPP
void output_restricted();
int main()
{
int choice;
clrscr();
printf("\n****** DEQUE ******");
printf("\n1. Input Restricted Deque");
printf("\n2. Output Restricted Deque");
printf("\nEnter your choice: ");
scanf("%d", &choice);
28:78
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

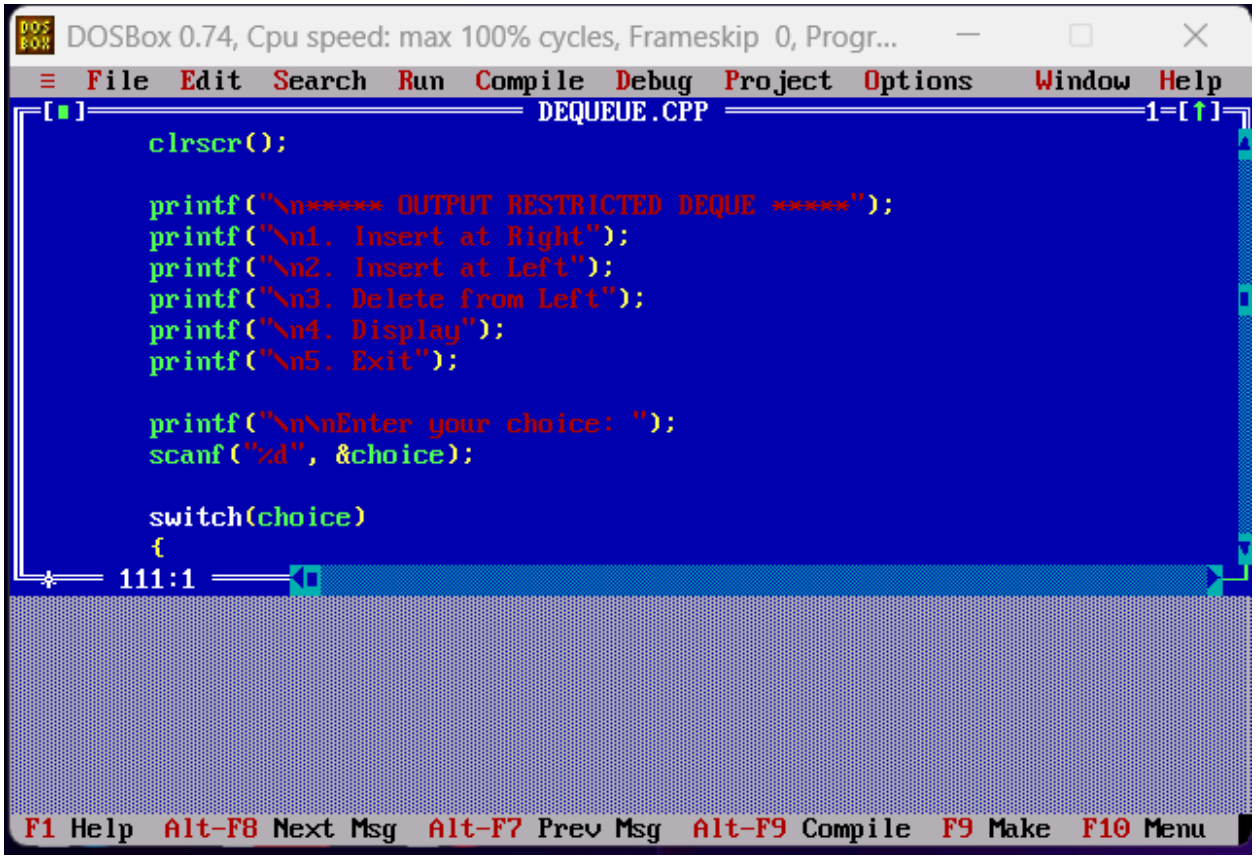

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
File Edit Search Run Compile Debug Project Options Window Help
DEQUEUE.CPP
1=[↑]
if (choice == 1)
input_restricted();
else if (choice == 2)
output_restricted();
else
printf("\nInvalid choice!");
getch();
return 0;
}
void input_restricted()
{
42:78
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

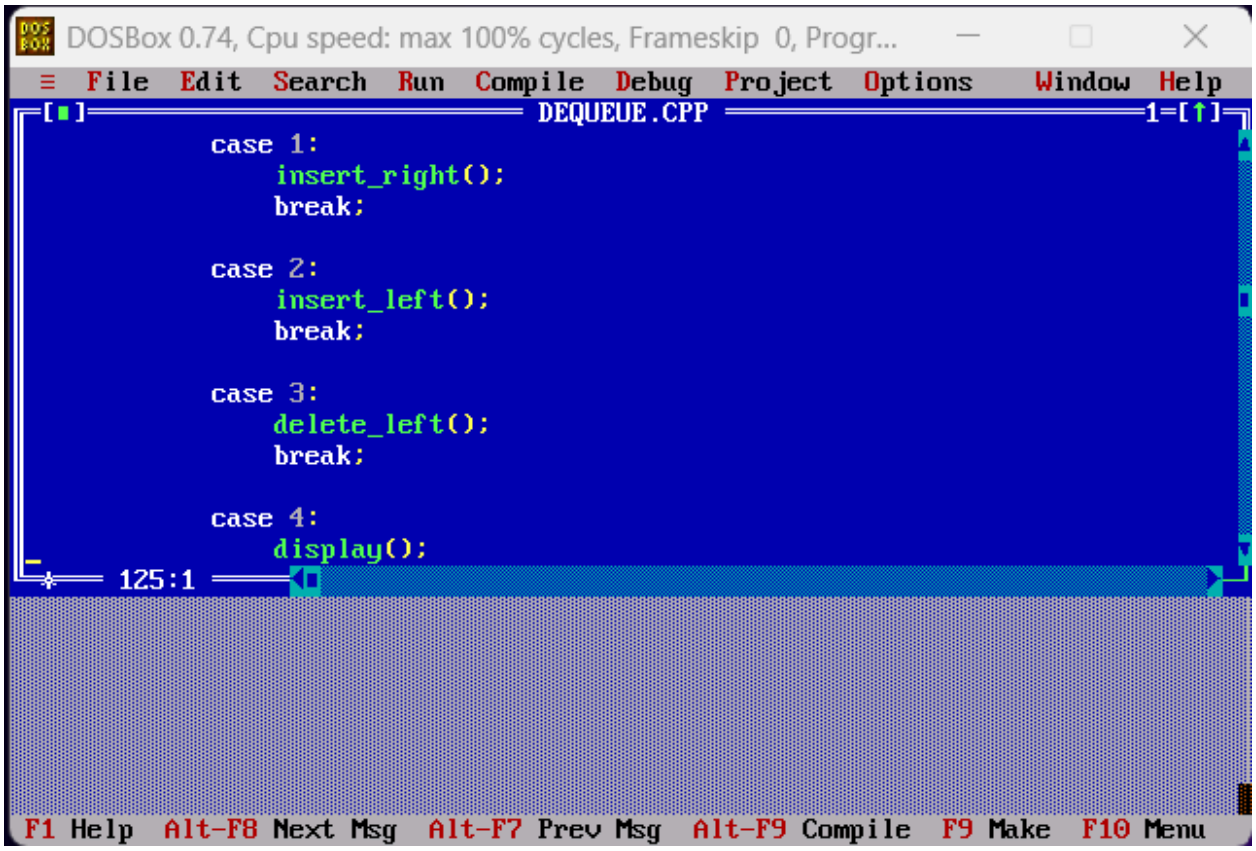

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
File Edit Search Run Compile Debug Project Options Window Help
DEQUEUE.CPP
1=[↑]
void input_restricted()
{
int choice;
do
{
clrscr();
printf("\n***** INPUT RESTRICTED DEQUE *****");
printf("\n1. Insert at Right");
printf("\n2. Delete from Left");
printf("\n3. Delete from Right");
printf("\n4. Display");
printf("\n5. Exit");
54:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

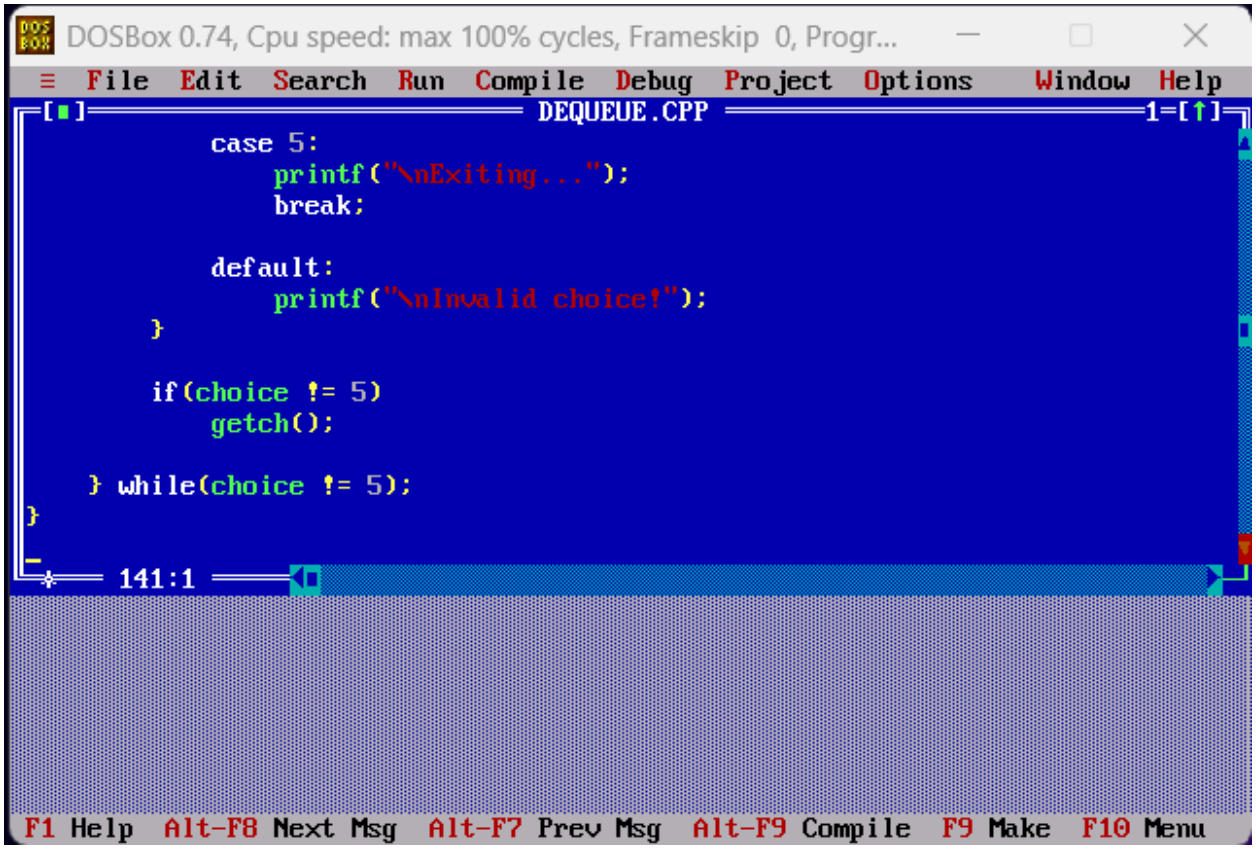

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
File Edit Search Run Compile Debug Project Options Window Help
DEQUEUE.CPP
printf("\n\nEnter your choice: ");
scanf("%d", &choice);
switch(choice)
{
case 1:
insert_right();
break;
case 2:
delete_left();
break;
case 3:
69:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

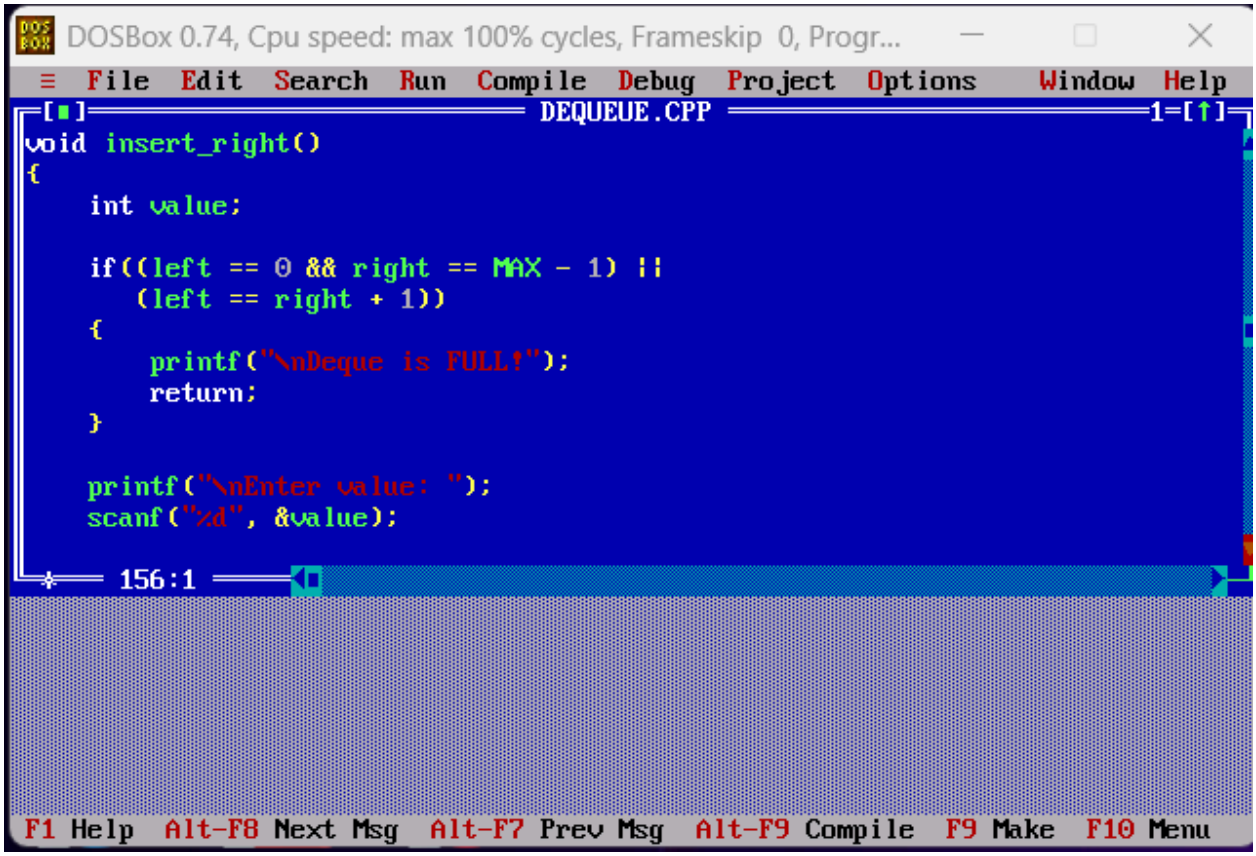

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
File Edit Search Run Compile Debug Project Options Window Help
DEQUEUE.CPP
case 3:
delete_right();
break;
case 4:
display();
break;
case 5:
printf("\nExiting...");
break;
default:
printf("\nInvalid choice!");
82:1

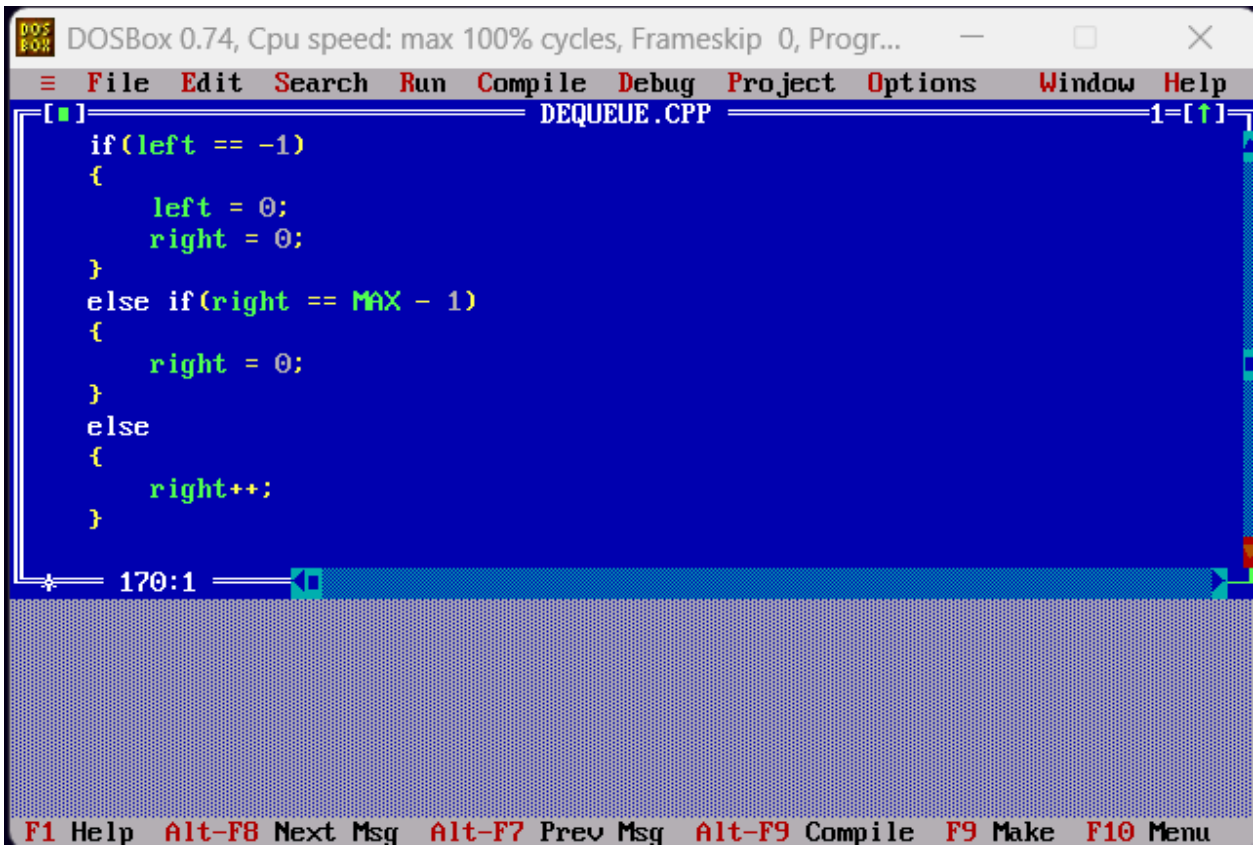

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
File Edit Search Run Compile Debug Project Options Window Help
DEQUEUE.CPP
}
if(choice != 5)
getch();
} while(choice != 5);
}
void output_restricted()
{
int choice;
do
96:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

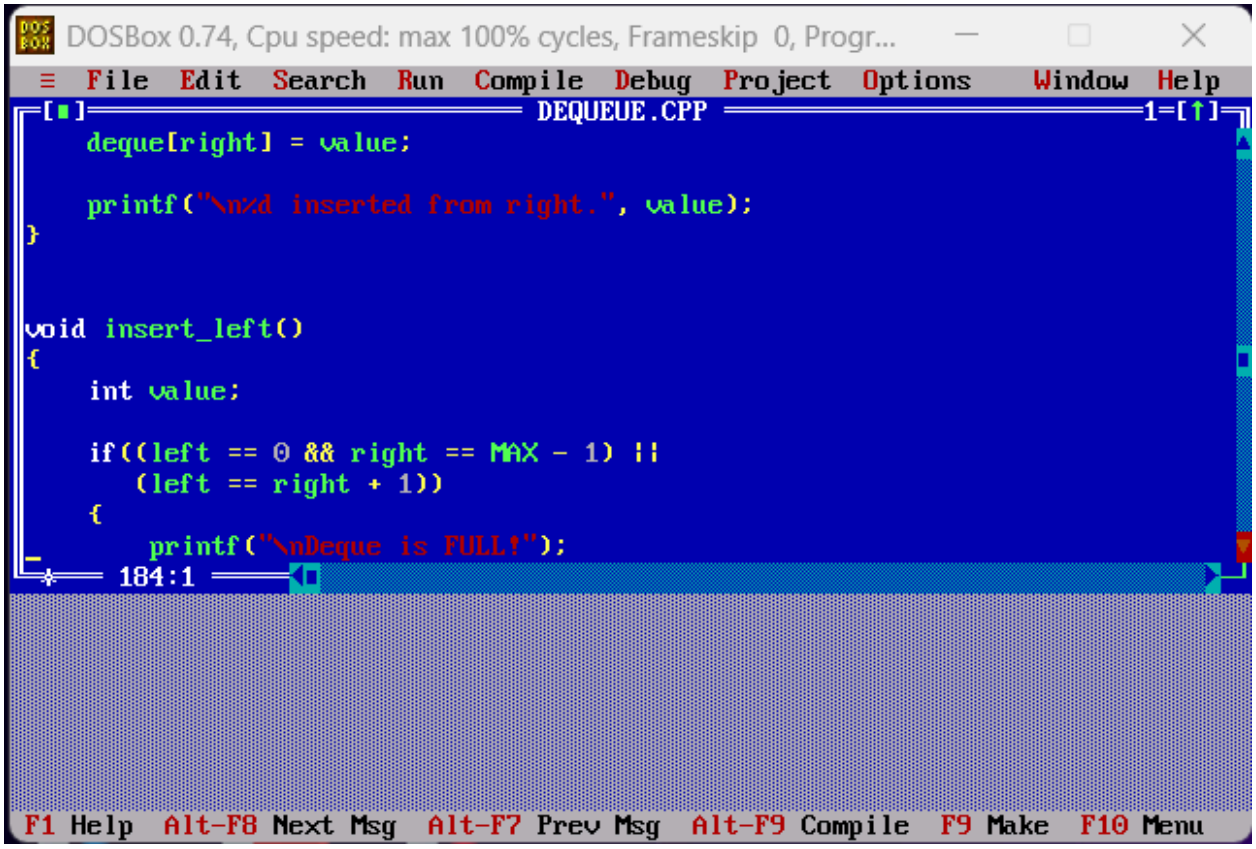

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
File Edit Search Run Compile Debug Project Options Window Help
DEQUEUE.CPP
clrscr();
printf("\n****** OUTPUT RESTRICTED DEQUE ******");
printf("\n1. Insert at Right");
printf("\n2. Insert at Left");
printf("\n3. Delete from Left");
printf("\n4. Display");
printf("\n5. Exit");
printf("\n\nEnter your choice: ");
scanf("%d", &choice);
switch(choice)
{
111:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

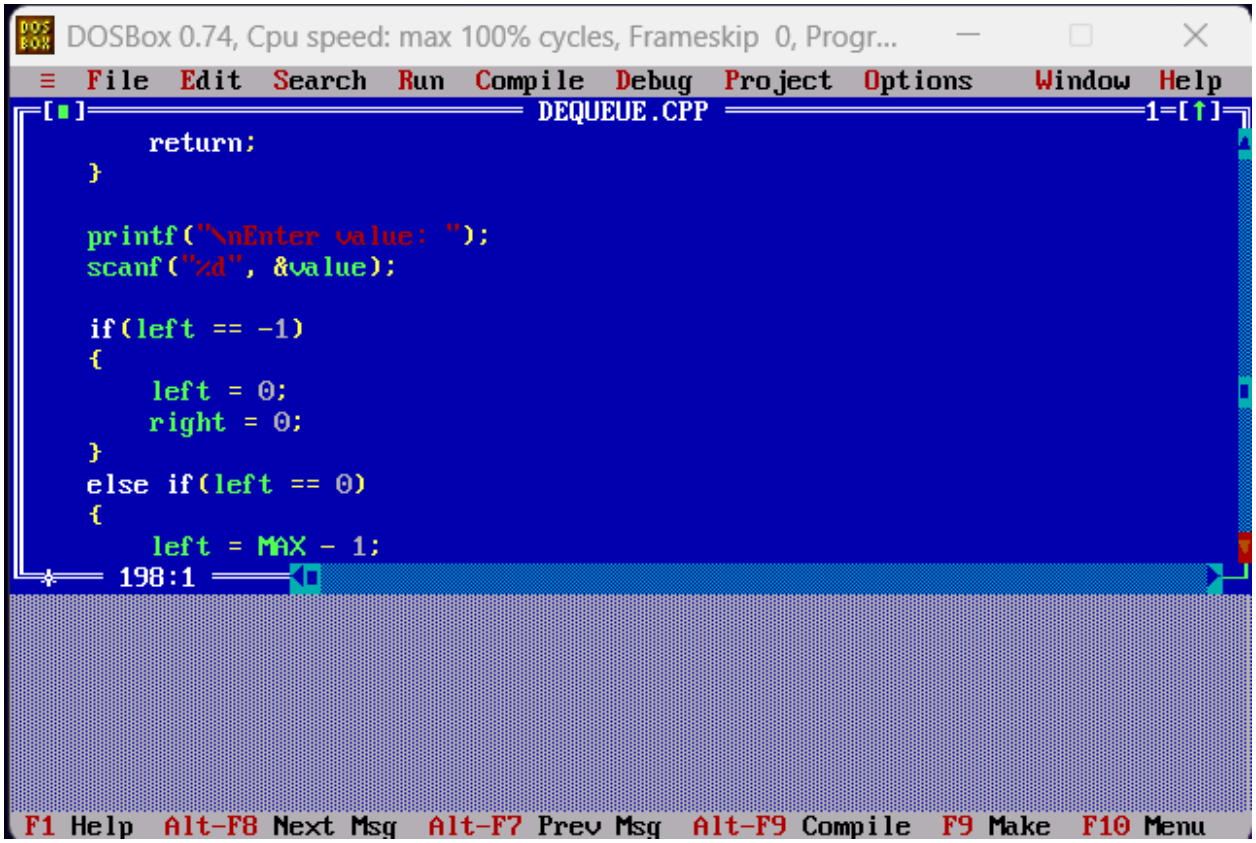

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
File Edit Search Run Compile Debug Project Options Window Help
DEQUEUE.CPP
case 1:
insert_right();
break;
case 2:
insert_left();
break;
case 3:
delete_left();
break;
case 4:
display();
125:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

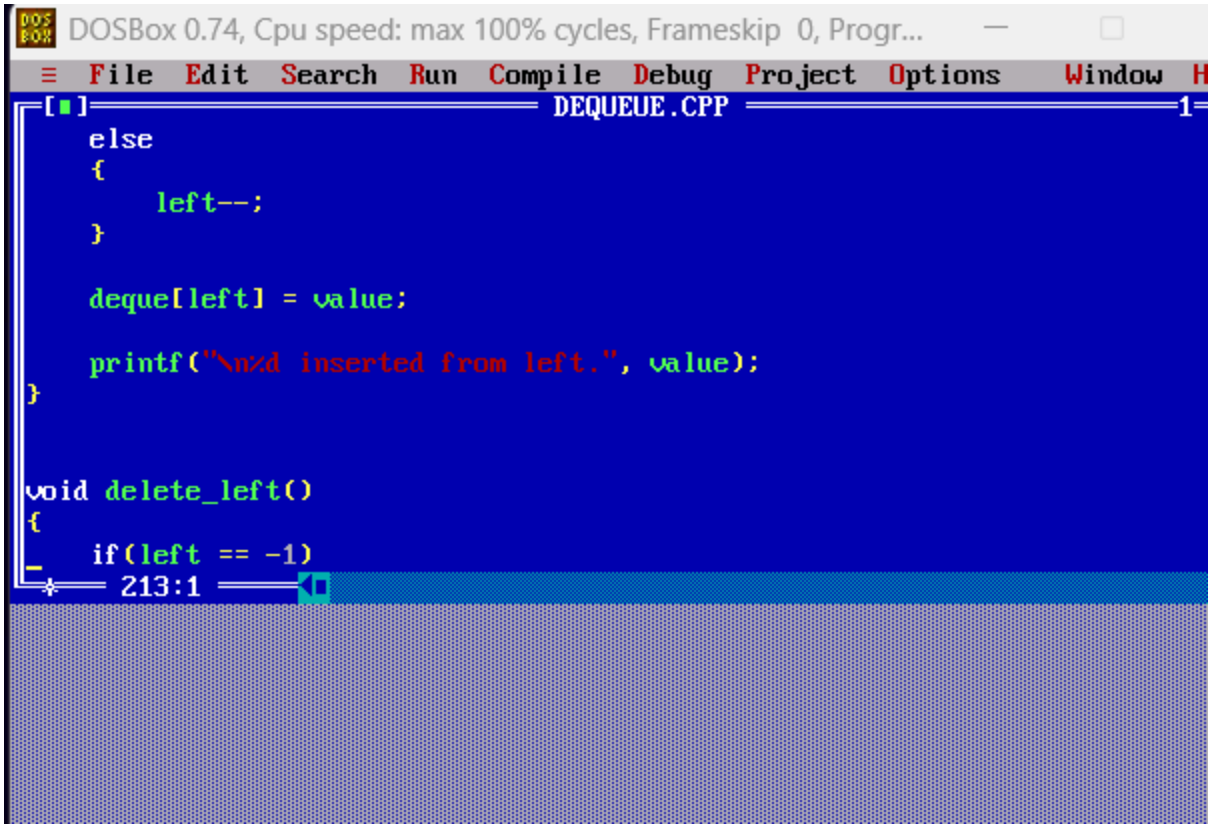

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
File Edit Search Run Compile Debug Project Options Window Help
DEQUEUE.CPP
case 5:
printf("\nExiting...");
break;
default:
printf("\nInvalid choice!");
}
if(choice != 5)
getch();
} while(choice != 5);
}
141:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

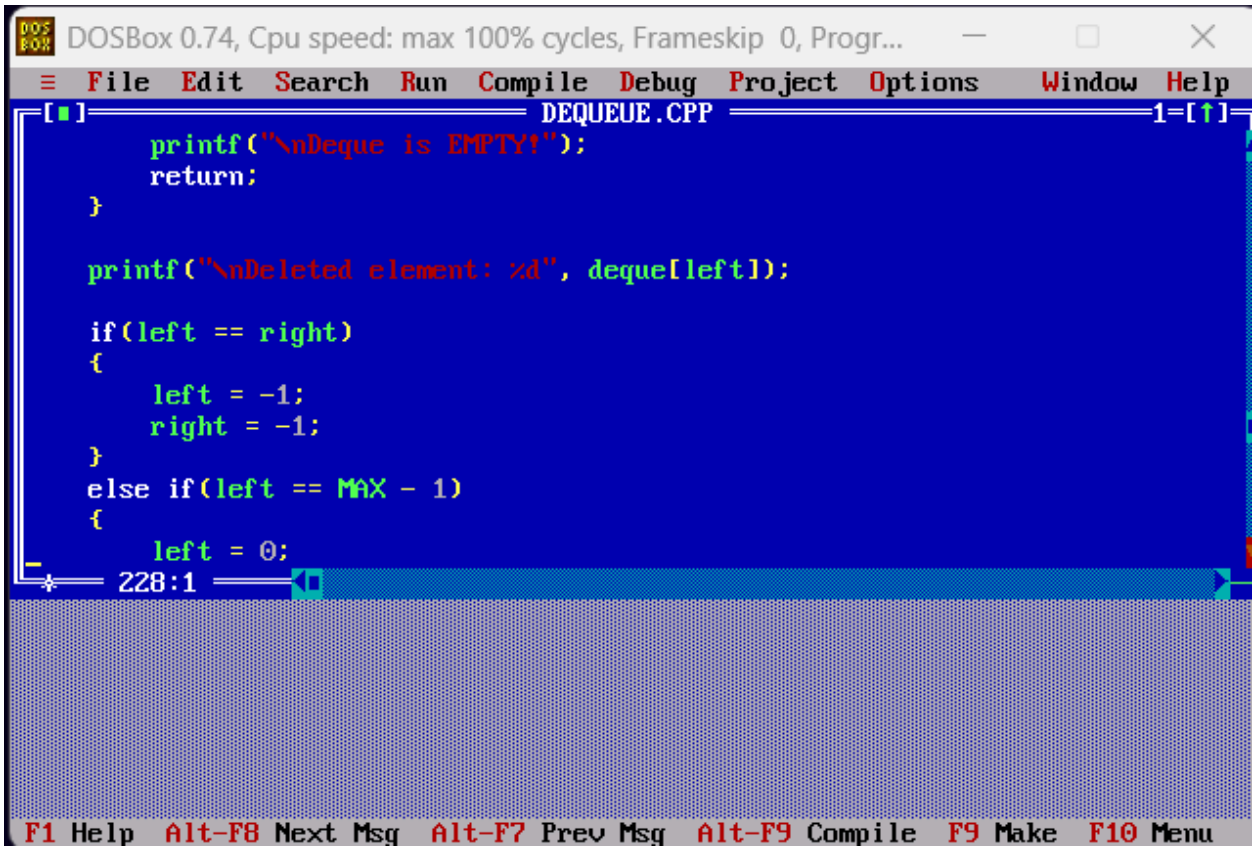

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
File Edit Search Run Compile Debug Project Options Window Help
DEQUEUE.CPP
1
void insert_right()
{
int value;
if((left == 0 && right == MAX - 1) ||
(left == right + 1))
{
printf("\nDeque is FULL!");
return;
}
printf("\nEnter value: ");
scanf("%d", &value);
156:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

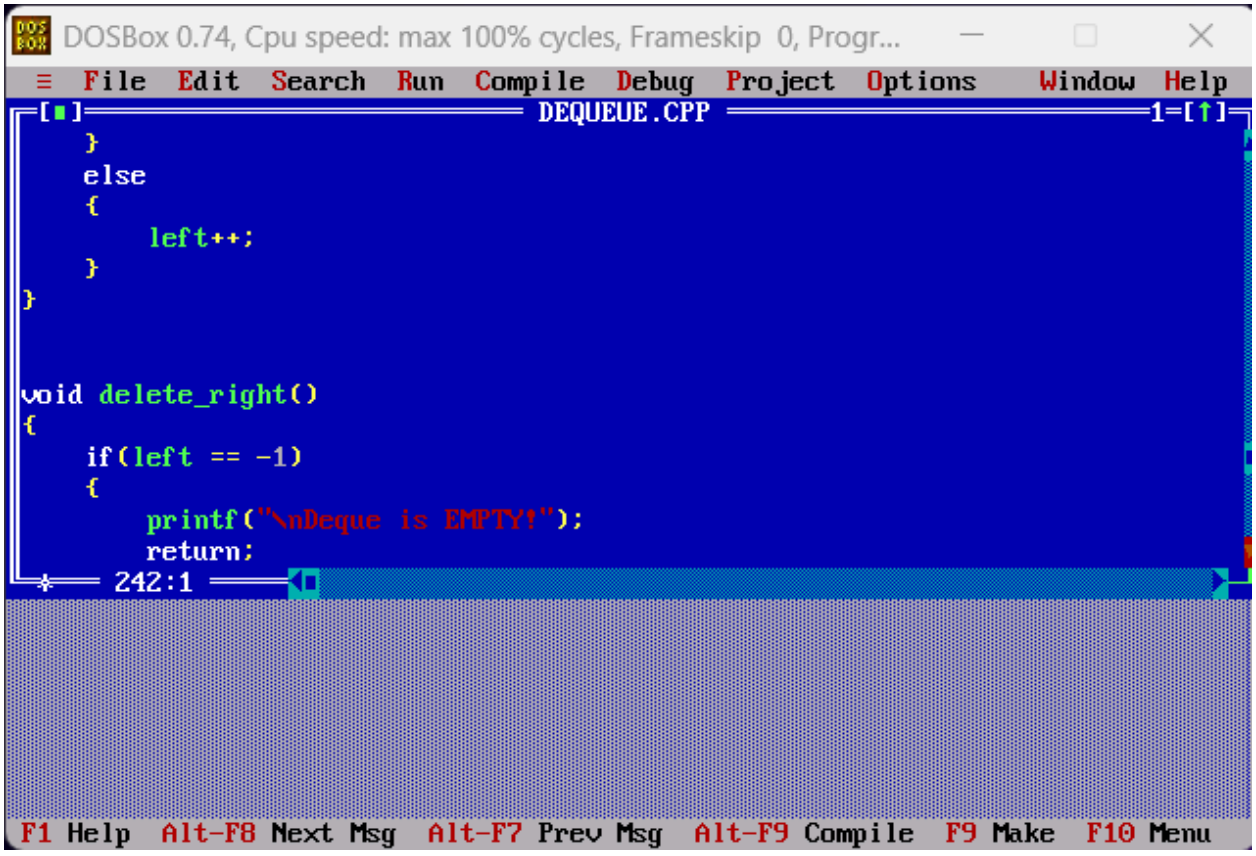

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
File Edit Search Run Compile Debug Project Options Window Help
DEQUEUE.CPP
1
if(left == -1)
{
left = 0;
right = 0;
}
else if(right == MAX - 1)
{
right = 0;
}
else
{
right++;
}
170:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

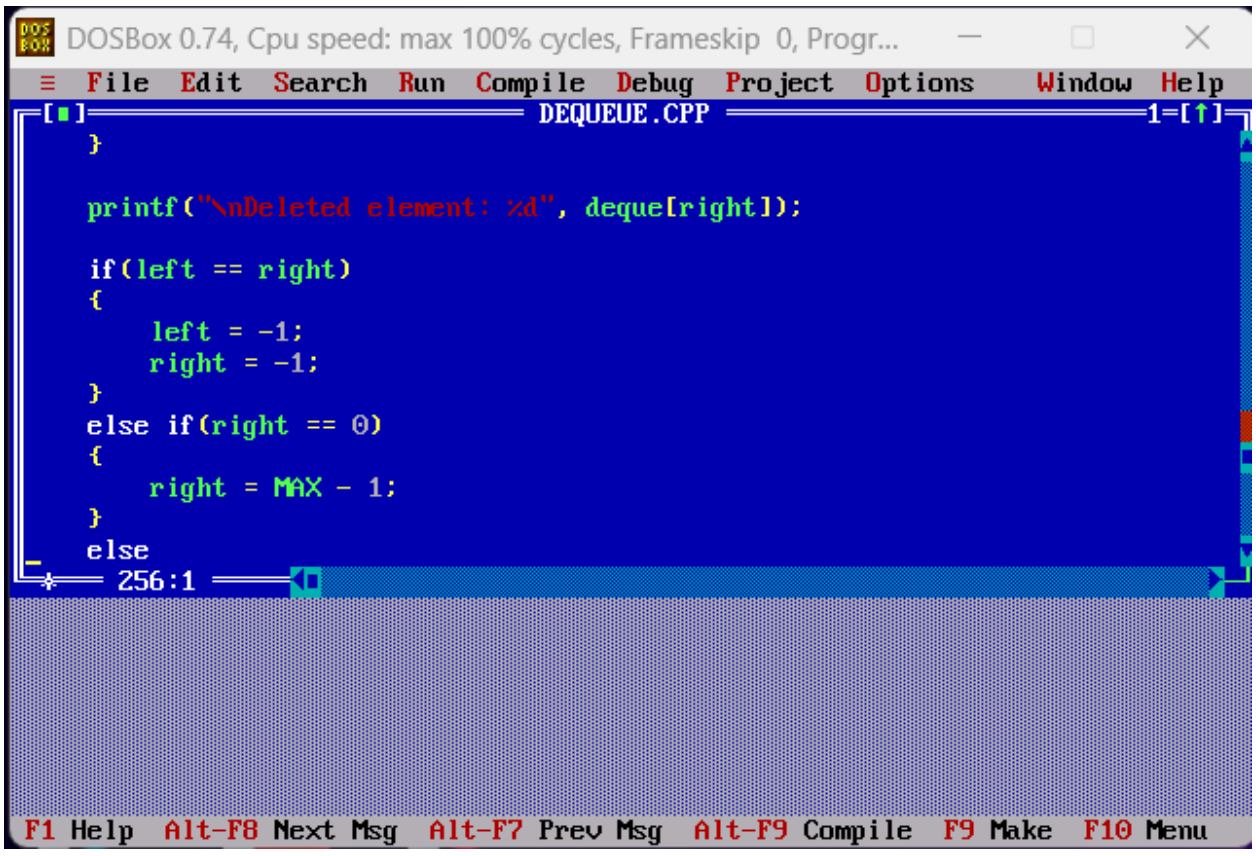

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
File Edit Search Run Compile Debug Project Options Window Help
DEQUEUE.CPP
deque[right] = value;
printf("\n%d inserted from right.", value);
}
void insert_left()
{
int value;
if((left == 0 && right == MAX - 1) ||
(left == right + 1))
{
printf("\nDeque is FULL!");
184:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

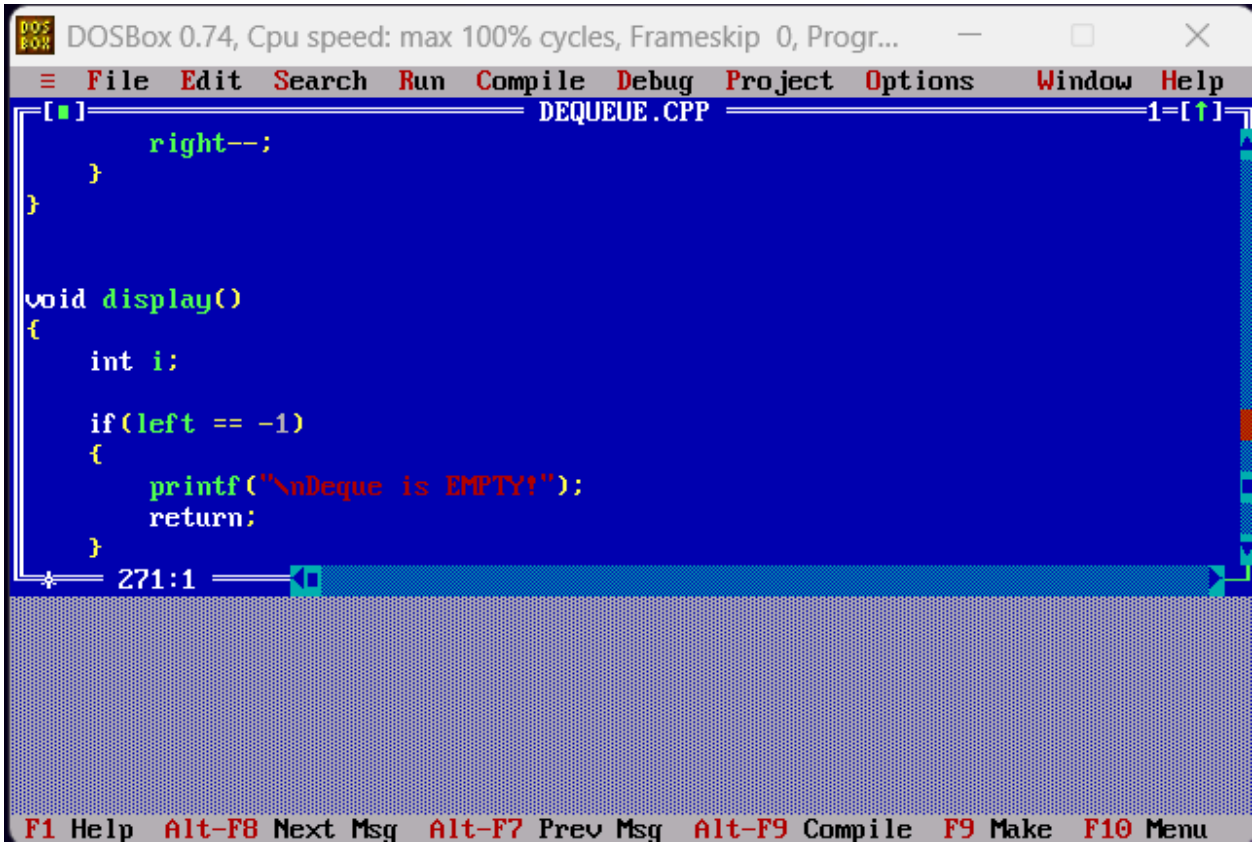

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
File Edit Search Run Compile Debug Project Options Window Help
DEQUEUE.CPP
return;
}
printf("\nEnter value: ");
scanf("%d", &value);
if(left == -1)
{
left = 0;
right = 0;
}
else if(left == 0)
{
left = MAX - 1;
198:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

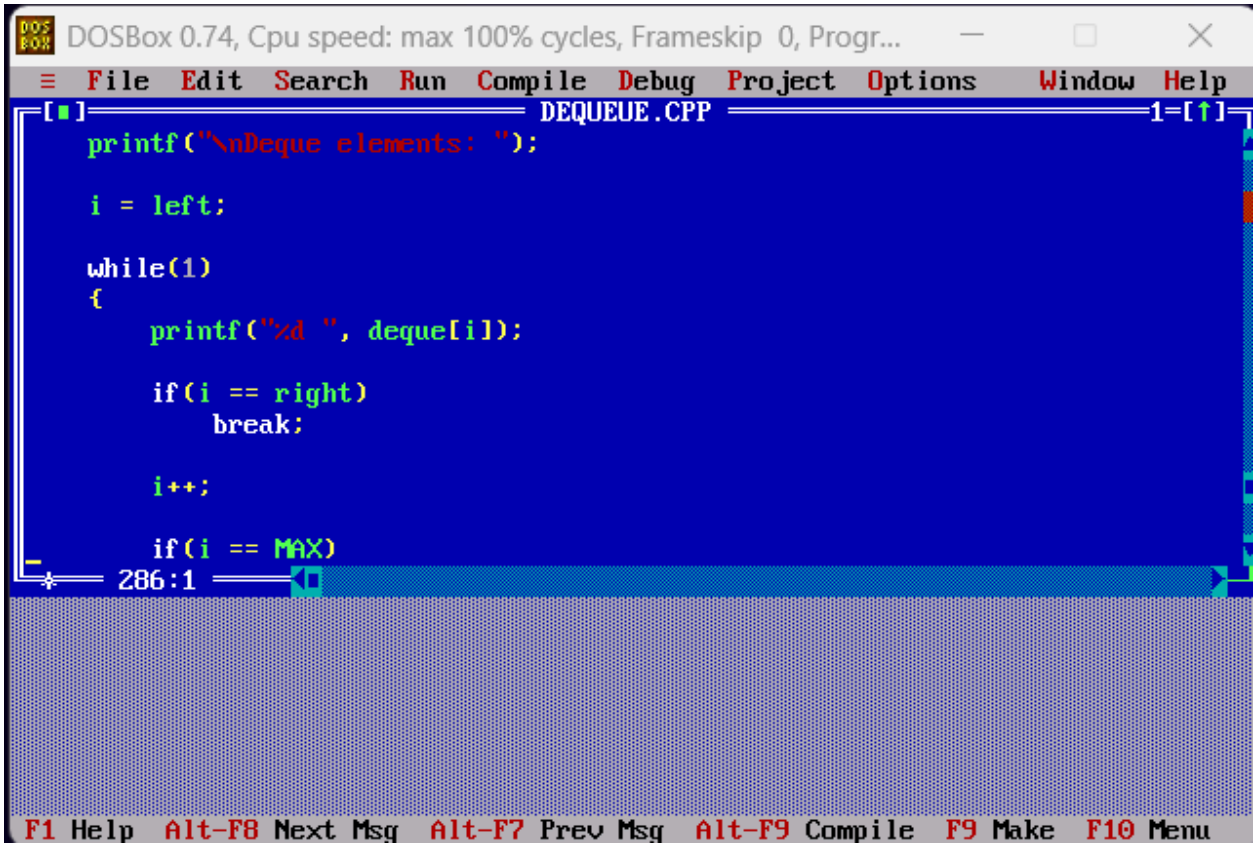

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
File Edit Search Run Compile Debug Project Options Window H
DEQUEUE.CPP
else
{
left--;
}
deque[left] = value;
printf("\n%d inserted from left.", value);
}
void delete_left()
{
if(left == -1)
213:1


DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
File Edit Search Run Compile Debug Project Options Window Help
DEQUEUE.CPP
printf("\nDeque is EMPTY!");
return;
}
printf("\nDeleted element: %d", deque[left]);
if(left == right)
{
left = -1;
right = -1;
}
else if(left == MAX - 1)
{
left = 0;
228:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

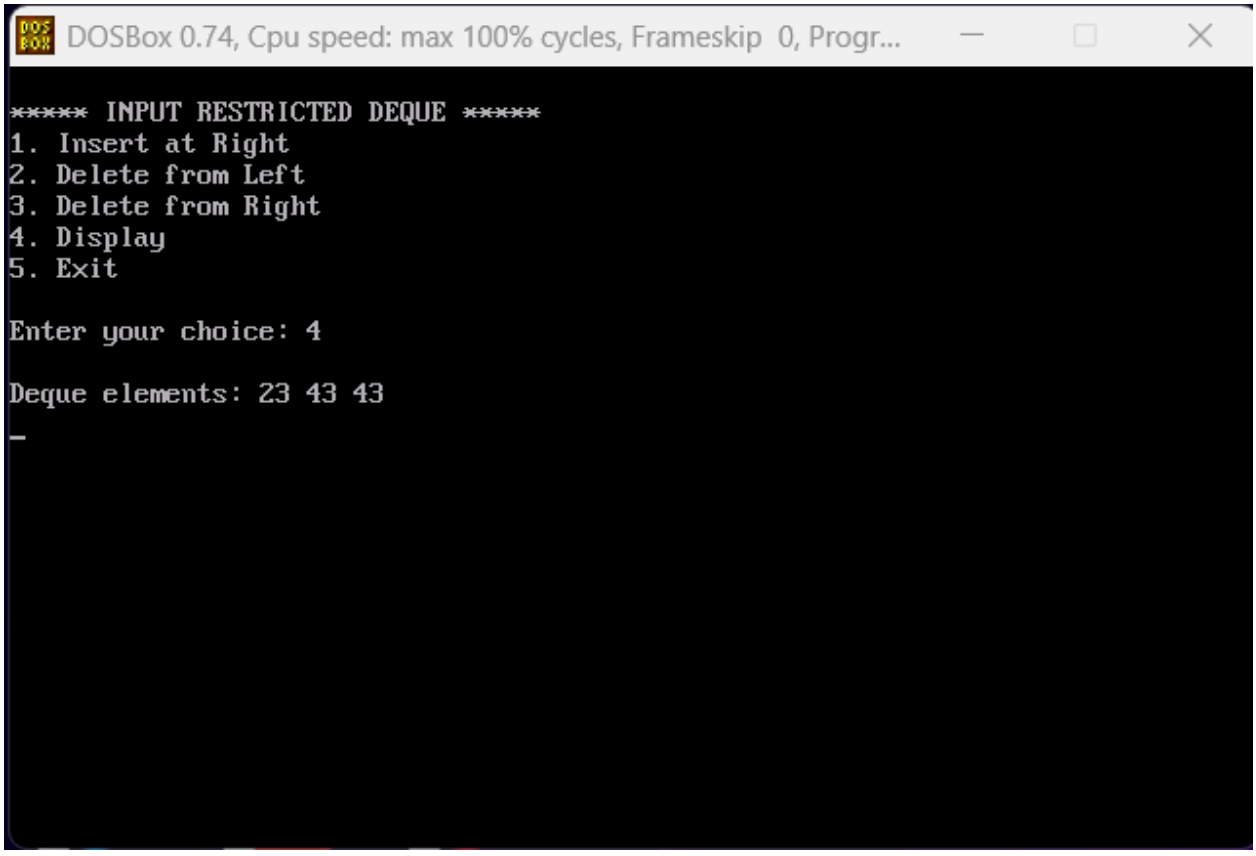

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
File Edit Search Run Compile Debug Project Options Window Help
DEQUEUE.CPP
}
else
{
left++;
}
}
void delete_right()
{
if(left == -1)
{
printf("\nDeque is EMPTY!");
return;
242:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

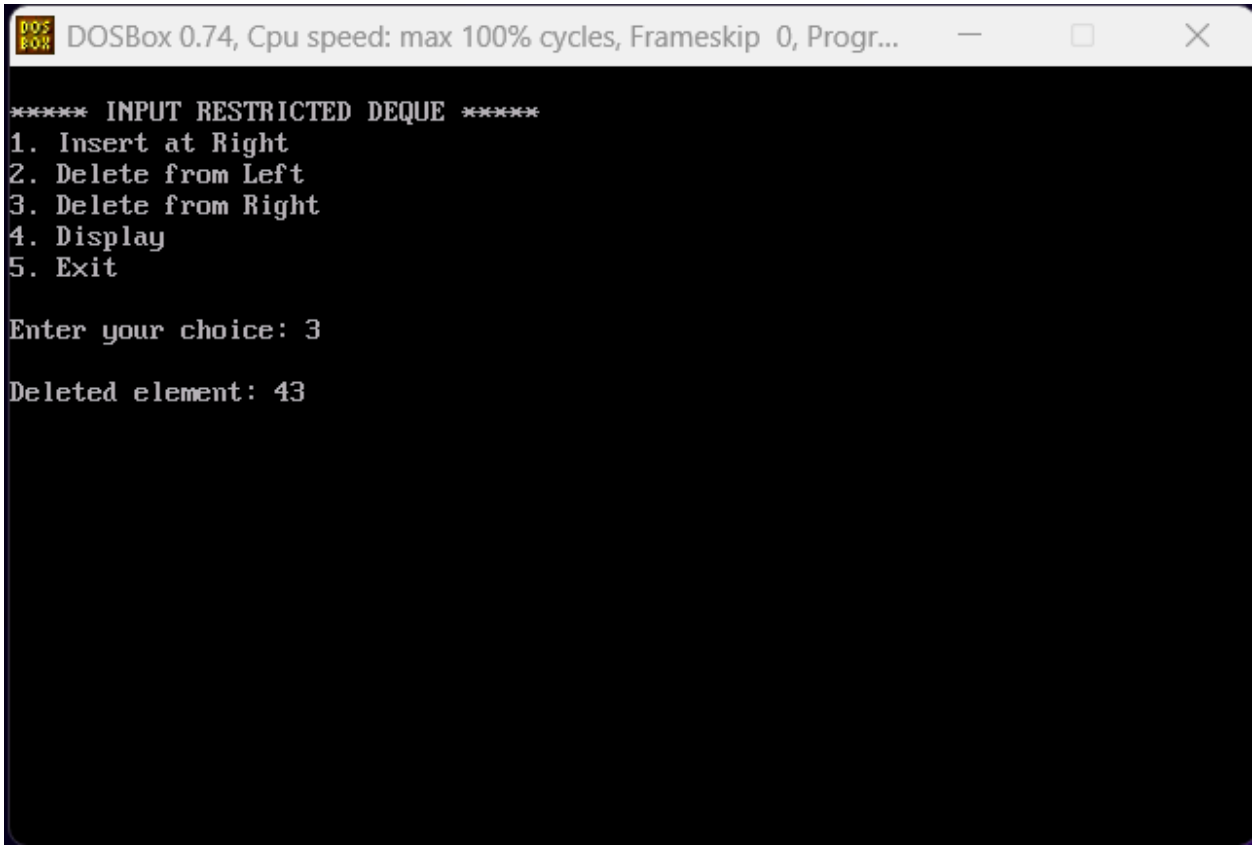

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
File Edit Search Run Compile Debug Project Options Window Help
DEQUEUE.CPP
}
printf("\nDeleted element: %d", deque[right]);
if(left == right)
{
left = -1;
right = -1;
}
else if(right == 0)
{
right = MAX - 1;
}
else
256:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

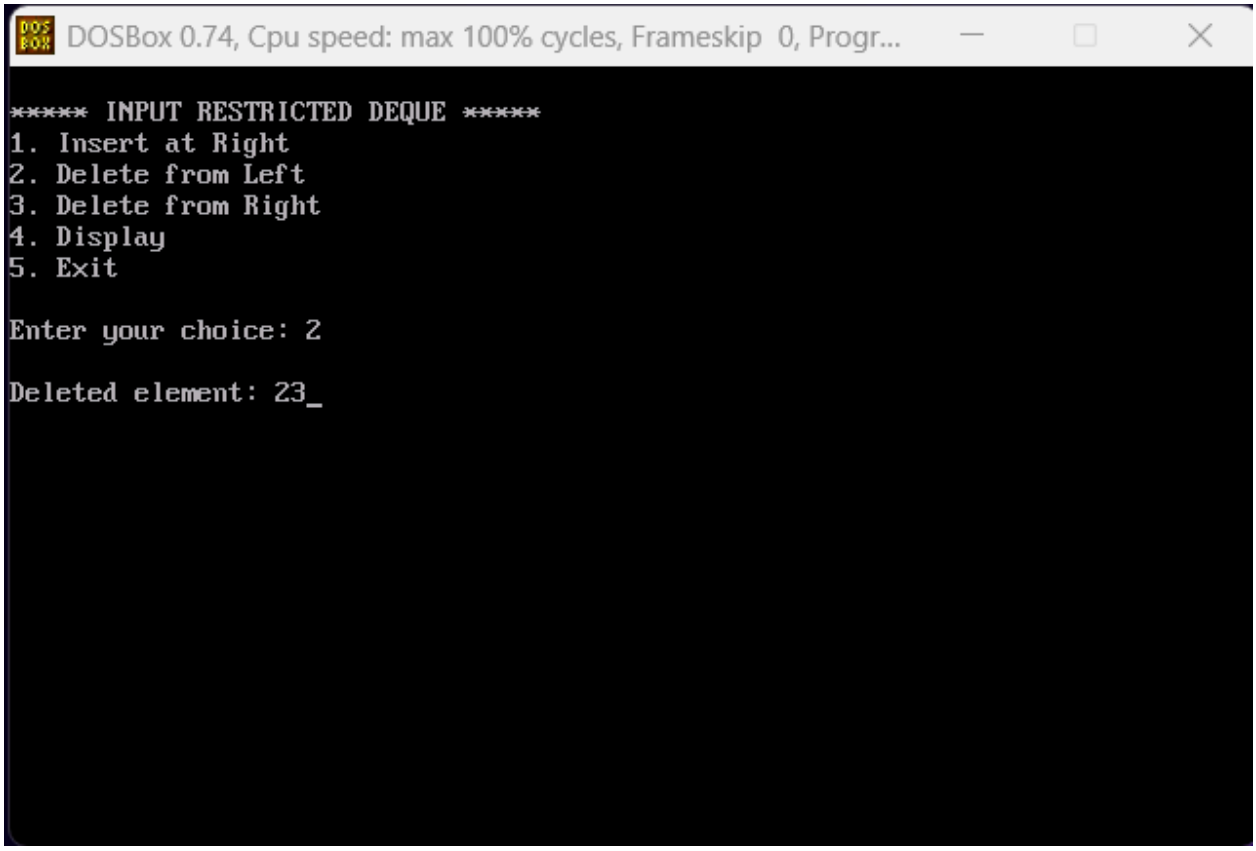

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
File Edit Search Run Compile Debug Project Options Window Help
DEQUEUE.CPP
1=[↑]
right--;
}
}
void display()
{
int i;
if(left == -1)
{
printf("\nDeque is EMPTY!");
return;
}
271:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

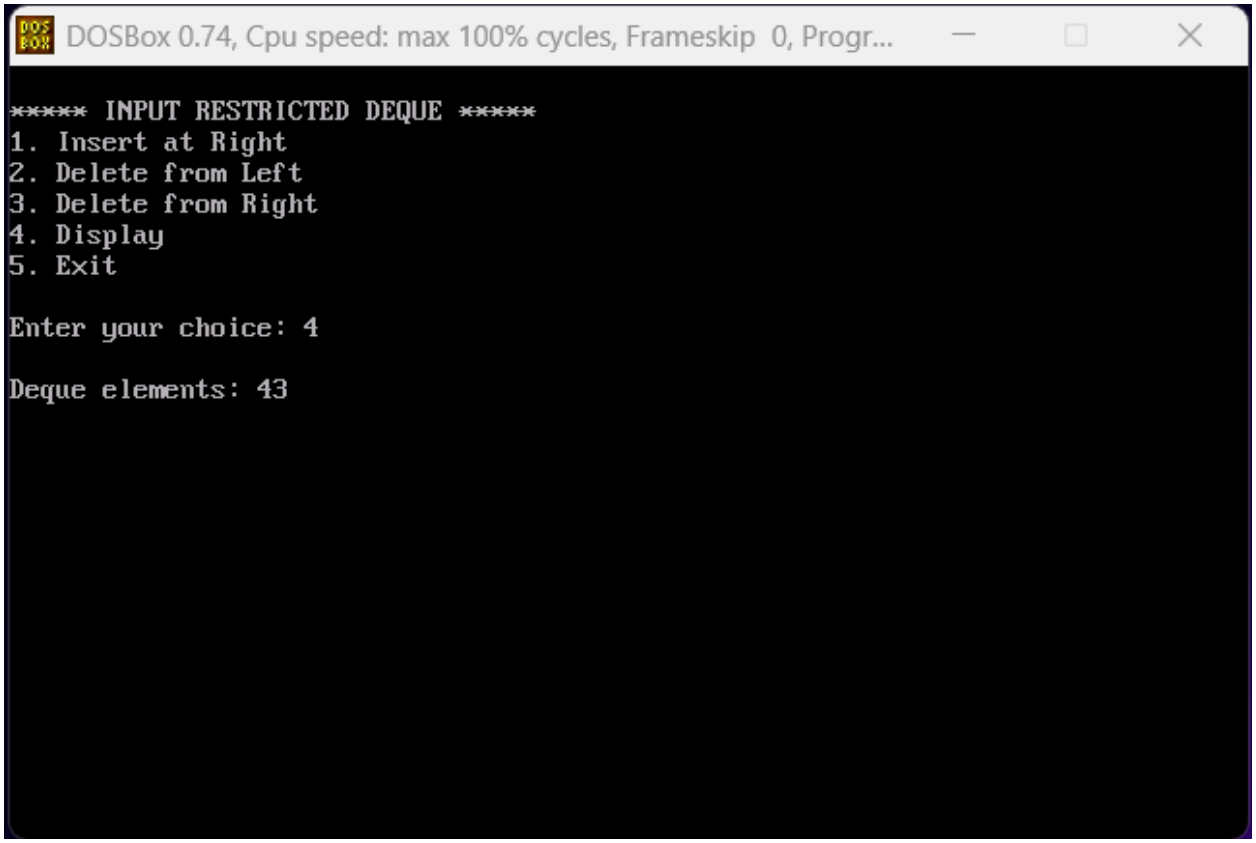

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
File Edit Search Run Compile Debug Project Options Window Help
DEQUEUE.CPP
1=[↑]
printf("\nDeque elements: ");
i = left;
while(1)
{
printf("%d ", deque[i]);
if(i == right)
break;
i++;
if(i == MAX)
286:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

output:


DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
***** INPUT RESTRICTED DEQUE *****
1. Insert at Right
2. Delete from Left
3. Delete from Right
4. Display
5. Exit
Enter your choice: 4
Deque elements: 23 43 43


DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
***** INPUT RESTRICTED DEQUE *****
1. Insert at Right
2. Delete from Left
3. Delete from Right
4. Display
5. Exit
Enter your choice: 3
Deleted element: 43


DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
***** INPUT RESTRICTED DEQUE *****
1. Insert at Right
2. Delete from Left
3. Delete from Right
4. Display
5. Exit
Enter your choice: 2
Deleted element: 23_


DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
***** INPUT RESTRICTED DEQUE *****
1. Insert at Right
2. Delete from Left
3. Delete from Right
4. Display
5. Exit
Enter your choice: 4
Deque elements: 43